

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

**Keep in Mind**

**This is a practice test.** It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. **100 points:** Part A 20, Part B 20, Part C 40, Part D 20.

**Part A — Vocabulary****10 questions, 2 points each**

Write the letter of the matching description next to each term.

- |                                   |                                                                          |
|-----------------------------------|--------------------------------------------------------------------------|
| 1. <b>D</b> Population            | A. A number that describes a sample.                                     |
| 2. <b>H</b> Sample                | B. The researcher assigns people to groups and gives them a treatment.   |
| 3. <b>G</b> Parameter             | C. A variable whose values are labels, like a color or a team name.      |
| 4. <b>A</b> Statistic             | D. The whole group we want to learn about.                               |
| 5. <b>C</b> Categorical variable  | E. Everyone has an equal chance of being picked.                         |
| 6. <b>I</b> Quantitative variable | F. The researcher watches or asks, without assigning anyone to a group.  |
| 7. <b>E</b> Simple random sample  | G. A number that describes a whole population.                           |
| 8. <b>J</b> Convenience sample    | H. The part of the population we actually collect data from.             |
| 9. <b>F</b> Observational study   | I. A variable whose values are measured amounts, like minutes or inches. |
| 10. <b>B</b> Experiment           | J. Choosing whoever is easiest to reach.                                 |

**Part B — Multiple Choice****5 questions, 4 points each**

11. Which of these variables is **quantitative**?
- (A) Eye color
  - (B) Favorite sport
  - (C) Height in inches ← **correct**
  - (D) Home ZIP code
12. Maplewood High School has 900 students. A survey of 60 of them finds that 30% walk to school. The 30% is a
- (A) statistic, because it describes the sample. ← **correct**
  - (B) parameter, because it describes the sample.
  - (C) statistic, because it describes all 900 students.
  - (D) parameter, because it describes all 900 students.
13. A librarian takes an alphabetical list of card holders, picks a random start, and then chooses **every 10th name**. This is a
- (A) stratified sample.
  - (B) cluster sample.
  - (C) convenience sample.
  - (D) systematic sample. ← **correct**

14. To learn how often town residents exercise, a researcher surveys people **inside a gym**. The results will most likely
- (A) be about right, because the people were chosen fairly.
  - (B) overestimate how often residents exercise. ← **correct**
  - (C) underestimate how often residents exercise.
  - (D) be right if the researcher surveys more people at the gym.
15. Researchers use a coin flip to put 80 volunteers into two groups. One group sleeps 8 hours and the other sleeps 5 hours before a memory test. This study is
- (A) an experiment, because the researchers assigned the groups. ← **correct**
  - (B) an experiment, because it measures memory.
  - (C) an observational study, because the volunteers were watched.
  - (D) an observational study, because there are two groups.

### Part C — Short Computations

4 questions, 10 points each

16. **Silver Lake Recreation Center** has 900 members. The director asks 60 members chosen at random, and 18 of them use the pool every week.
- (a) What percent of the sample uses the pool every week?

$$\frac{18}{60} = 0.30 = 30\%$$

- (b) About how many of all 900 members use the pool every week?

$$0.30 \times 900 = 270 \text{ members}$$

17. **Granite Falls Middle School** has 600 students and wants a **stratified** sample of 60 — the same fraction from each grade.
- (a) What fraction of the school is being sampled?

$$\frac{60}{600} = \frac{1}{10}$$

- (b) Complete the table.

| Grade             | 6   | 7   | 8   | Total |
|-------------------|-----|-----|-----|-------|
| Students in grade | 240 | 200 | 160 | 600   |
| Number sampled    | 24  | 20  | 16  | 60    |

18. **Brookside Apartments** has 500 households on a numbered list. The manager wants a **systematic** sample of 50.
- (a) Find the interval  $k$ .

$$k = 500 \div 50 = 10$$

- (b) The random start is household 3. List the first four households chosen.

3   13   23   33

19. In an **experiment**, 40 students are randomly assigned to use a flashcard app and 40 to study the usual way. Afterwards, 30 of the app group and 20 of the usual group passed a quiz.
- (a) Find the percent who passed in each group.

**App:**  $\frac{30}{40} = 0.75 = 75\%$

**Usual:**  $\frac{20}{40} = 0.50 = 50\%$

- (b) How many percentage points apart are the two groups? **25 points**

#### Part D — Critical Thinking

1 question, 20 points

20. A random sample of Maplewood students found that students who **eat breakfast** have higher grades than students who skip it. Nobody was assigned — students chose for themselves. The principal says, “*Eating breakfast causes higher grades.*”

Is the principal’s conclusion justified? Explain your reasoning, and name one **other variable** that could explain the result.

**No. Nobody was assigned — students chose — so this is an observational study.** .....

**It shows an association between breakfast and grades, not causation.** .....

**Another variable: family routine. Students with more support at home are more** .....

**likely to eat breakfast *and* to keep up with homework,** .....

**and that support could raise grades by itself.** .....

**Only an experiment that randomly assigns breakfast could show it causes higher grades.** .....